CASE STUDY 47

SPORTS TOURNAMENT RANKING SYSTEM

Python Program:

```
print("===== SPORTS TOURNAMENT RANKING SYSTEM =====")


teams = []

points_list = []


n = int(input("Enter the number of teams: "))


if n <= 0:

   print("Error: Number of teams must be positive.")


else:

   for i in range(n):

      team_name = input("Enter team {} name: ".format(i + 1))

      points = int(

         input("Enter points for {}: ".format(team_name))

      )


      teams.append(team_name)

      points_list.append(points)


   # Combine team names and points
```

```
tournament = list(zip(teams, points_list))

# Sort teams by points in descending order
tournament.sort(key=lambda item: item[1], reverse=True)

print("\n===== FINAL TOURNAMENT RANKINGS =====")

for rank, (team, points) in enumerate(tournament, start=1):
    print("Rank {}: {} - {} points".format(rank, team, points))

print("\n===== TOURNAMENT SUMMARY =====")

highest_points = tournament[0][1]
winners = [
    team for team, points in tournament
    if points == highest_points
]

if len(winners) == 1:
    print("Tournament Winner:", winners[0])
else:
    print("Joint Winners:", ", ".join(winners))

print("Highest Points:", highest_points)
print("Total Teams:", n)
```

```
print("\nRanking completed.")
```

Output:

